Learning Agile Locomotion on Risky Terrains

Chong Zhang[1], Nikita Rudin[1], David Hoeller[1], Marco Hutter[1]

***Abstract*— Quadruped robots have shown remarkable mobility on various terrains through reinforcement learning. Yet, in the presence of sparse footholds and risky terrains such as stepping stones and balance beams, which require precise foot placement to avoid falls, model-based approaches are often used. In this paper, we show that end-to-end reinforcement learning can also enable the robot to traverse risky terrains with dynamic motions. To this end, our approach involves training a generalist policy for agile locomotion on disorderly and sparse stepping stones before transferring its reusable knowledge to various more challenging terrains by finetuning specialist policies from it. Given that the robot needs to rapidly adapt its velocity on these terrains, we formulate the task as a navigation task instead of the commonly used velocity tracking which constrains the robot's behavior and propose an exploration strategy to overcome sparse rewards and achieve high robustness. We validate our proposed method through simulation and real-world experiments on an ANYmal-D robot achieving peak forward velocity of ≥ 2.5 m/s on sparse stepping stones and narrow balance beams. Video: youtu.be/Z5X0J8OH6z4**

I. INTRODUCTION

Quadrupedal robots require perceptive locomotion abilities on rough and complex terrains, including risky terrains with sparse footholds such as stepping stones and balance beams. Existing works typically employ model-based controllers [1]–[6] which perform well in simulation or under laboratory conditions. However, they are empirically difficult to deploy in the wild due to model mismatch and unexpected slippage [7], [8], and typically require dedicated design (e.g. motion references [9] and handcrafted heuristics such as terrain roughness and pose safety [3]) to achieve specified motions. Furthermore, the dynamics model needs to be simplified especially for the limbs [2], [4], and replanning is often slow due to the heavy computational burden [3], [5].

On the other hand, model-free reinforcement learning (RL)-based methods demonstrate excellent robustness in the wild via domain randomization and massive training data [10], [11]. However, no existing works have used end-to-end RL to handle the sparse footholds on highly risky terrains, where precise motions are necessary to avoid failures, and exploration is difficult due to the numerous ways to fail. First attempts to solve such locomotion challenges using end-to-end learning show promising results, but still have limited terrain complexity [12] or need to exactly know which stones to step on [13]. Another work trains decoupled policies for foothold planning and execution [14], but it cannot achieve dynamic locomotion on risky terrains, and

[1] Authors are with the Robotic Systems Lab, ETH Zurich, 8092 Zurich, Switzerland. Emails: {chozhang, rudinn, dhoeller, mahutter}@ethz.ch

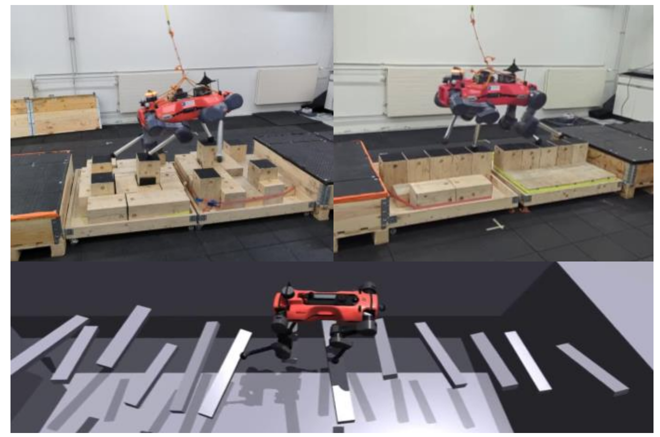

Fig. 1. The ANYmal-D robot executing the learned RL policies on stepping stones, balance beams with gaps, and stepping beams. **Left top:** the stepping stones were 20 cm wide taking up only $\sim 10\%$ of the whole area, and the height difference between high and low stones was 12 cm. The robot reached the peak forward velocity of 2.7 m/s when traversing the stones from left to right. **Right top:** the terrain consisted of two beams of 100 cm $\times$ 20 cm and three gaps. The peak forward velocity of the robot was 2.5 m/s from right to left. **Bottom:** The widths of stepping beams varied in [12 cm, 17 cm], the horizontal distances between two beams varied in [30 cm, 60 cm], and the vertical distances varied in [0 cm, 20 cm]. For practical reasons, we only tested the policy in simulation, and the peak forward velocity of the robot reached 1.5 m/s from left to right.

decoupled planning and control can inherently limit the range of behaviors that can be expressed [15].

Some existing works combine RL and model-based controllers. For example, [16] uses RL to generate desired foot positions which are then tracked by a whole-body motion controller, as well as corrective joint torques to achieve domain adaptative tracking. Another work uses RL to generate the desired acceleration of the base and tracks it with model-based controllers [17]. All of these approaches, however, demonstrate limited agility with velocity ≤ 0.5m/s. Another recent work [18] opens a new venue by tracking trajectories from a model-based planner with an RL policy and demonstrates remarkable performance, but the motions are constrained by the planner and cannot recover from failures.

In this paper, we explore how dynamic locomotion of a quadrupedal robot can be achieved on highly risky terrains through end-to-end RL. We first train a generalist policy to learn reusable sensorimotor skills on sparse and disorderly stepping stones. Then, we finetune multiple specialist policies from the generalist policy on various more challenging terrains, and successfully transfer two of the specialist policies to real-world stepping stones and narrow beams. We achieve highly agile motions with peak forward velocity ≥ 2.5 m/s, as shown in Fig. 1.

在危险地形上学习敏捷运动

张翀[1], Nikita Rudin[1],David Hoeller[1],Marco Hutter[1]

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摘要——**四足机器人通过强化学习已在各种地形上展现出卓越的移动能力。然而，在存在稀疏立足点和危险地形（如踏脚石和平衡木）的情况下，这些地形需要精确的落脚点以避免摔倒，通常采用基于模型的方法。在本文中，我们表明端到端强化学习也能使机器人以动态运动穿越危险地形。为此，我们的方法包括先在杂乱且稀疏的踏脚石上训练一个用于敏捷运动的通用策略，然后通过从中微调专家策略，将其可复用知识迁移到各种更具挑战性的地形上。鉴于机器人需要在这些地形上快速调整其速度，我们将任务表述为导航任务，而非常用的会约束机器人行为的速度跟踪，并提出一种探索策略来克服稀疏奖励并实现高鲁棒性。我们通过在ANYmal-D机器人上进行的仿真和真实世界实验验证了所提出的方法，在稀疏踏脚石和狭窄平衡木上达到了 ≥ 2.5 米/秒的峰值前进速度。视频：youtu.be/Z5X0J8OH6z4**

I. 引言

四足机器人需要在崎岖复杂的地形上具备感知运动能力，包括具有稀疏立足点的危险地形，如踏脚石和平衡木。现有工作通常采用基于模型的控制器[1]–[6]，这些控制器在仿真或实验室条件下表现良好。然而，由于模型失配和意外滑移[7], [8], ，它们在实际野外部署中经验上较为困难，并且通常需要专门设计（例如运动参考 [9] 以及诸如地形粗糙度和姿态安全 [3]等手工启发式规则）来实现特定运动。此外，动力学模型需要简化，尤其是对于肢体 [2], [4], ，而且由于计算负担沉重，重规划往往很慢 [3], [5]。

另一方面，基于无模型强化学习（RL）的方法通过域随机化和大量训练数据在野外展现出卓越的鲁棒性 [10], [11]。然而，现有工作尚未使用端到端强化学习来处理高度危险地形上的稀疏立足点，在这些地形中，精确的动作对于避免失败至关重要，而由于失败方式众多，探索也颇具挑战。首次尝试使用端到端学习解决此类运动挑战的方法显示出令人鼓舞的结果，但仍受限于地形复杂度 [12]或需要精确知道应踩踏哪些石头 [13]。另一项工作为立足点规划与执行训练了解耦策略 [14], ，但该方法无法在危险地形上实现动态运动，且

[1] 作者隶属于苏黎世联邦理工学院机器人系统实验室，瑞士苏黎世8092。邮箱： {chozhang, rudinn, dhoeller, mahutter}@ethz.ch

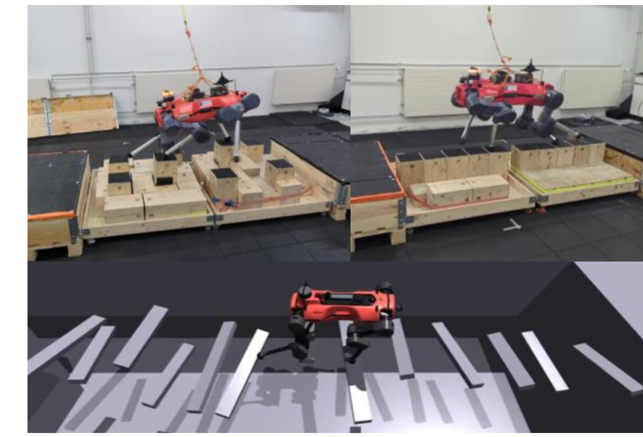

图1. ANYmal-D 机器人在踏脚石、带间隙的平衡木和踏脚梁上执行学习到的强化学习策略。**左上：**踏脚石宽20厘米，仅占整个区域的 ~ 10 %，高低石头之间的高度差为12厘米。机器人从左到右穿越石头时达到了2.7米/秒的前进速度峰值。**右上：**地形由两根100厘米 $\times$ 20 厘米的横梁和三个间隙组成。机器人从右到左的前进速度峰值为2.5米/秒。**底部：**踏脚梁的宽度在 [12 厘米,17厘米], 之间变化，两根梁之间的水平距离在 [30 厘米,60厘米],之间变化，垂直距离在 [0 厘米,20厘米]之间变化。出于实际原因，我们仅在仿真中测试了该策略，机器人从左到右的前进速度峰值达到了1.5米/秒。

解耦规划与控制本质上会限制可表达的行为范围 [15]。

一些现有工作将强化学习与基于模型的控制器相结合。例如， [16] 使用强化学习生成期望的足部位置，随后由全身运动控制器进行跟踪，并辅以校正关节力矩以实现域自适应跟踪。另一项工作使用强化学习生成基座的期望加速度，并通过基于模型的控制器 [17]进行跟踪。然而，所有这些方法在速度 ≤ 0.5m/s下展现的敏捷性有限。另一项近期工作 [18] 开辟了新途径，通过强化学习策略跟踪基于模型的规划器生成的轨迹，并展现出卓越的性能，但这些运动受限于规划器，且无法从故障中恢复。

在本文中，我们探讨了如何通过端到端强化学习在高度危险的地形上实现四足机器人的动态运动。我们首先训练一个通用策略，使其在稀疏且无序的踏脚石上学习可复用的感觉运动技能。然后，我们从通用策略出发，在各种更具挑战性的地形上微调多个专家策略，并成功将其中两个专家策略迁移到真实世界的踏脚石和窄梁上。我们实现了峰值前进速度≥ 2.5 米/秒的高度敏捷动作，如图1所示。

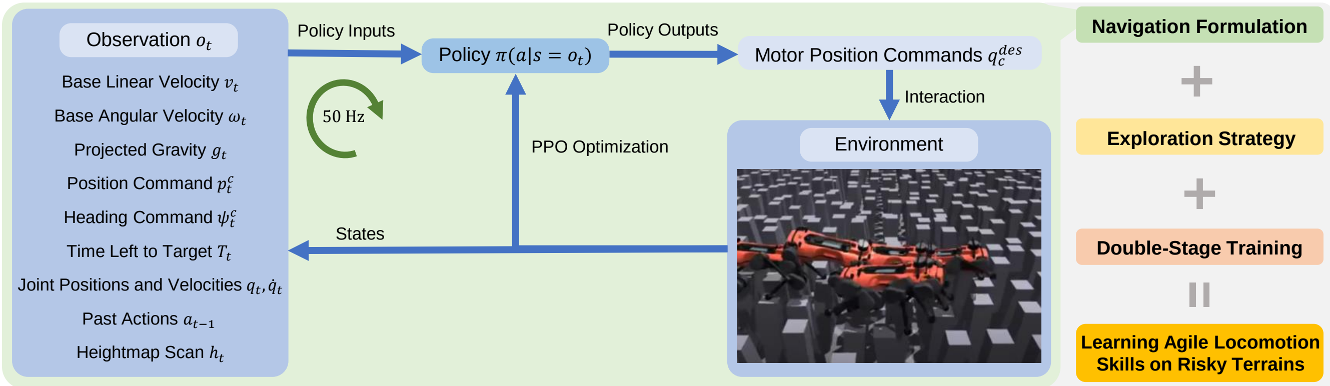

Fig. 2. An overview of our methodology. The learning system is illustrated on the left. Different components are listed on the right.

Traditional learning methods based on the commonly used velocity tracking formulation (e.g., the one used in [19] and [12]) can not achieve these challenging tasks. It is intractable to assign appropriate velocity commands to the robots on risky terrains especially when they need to rapidly adapt their velocities, and the feasible solutions are quite rare and hard to explore compared to nonrisky terrains. Hence, the key idea in this paper is to formulate the task as a navigation problem as in [20], though the risky environments and the sparse rewards make the exploration even harder.

To overcome the hard exploration in the navigation formulation, we improve the curriculum learning design in [20], and incorporate curiosity-driven intrinsic rewards [21]. We also employ a symmetry-based data augmentation from [22], utilizing the left-right and front-back symmetry of our robot to improve the data efficiency. These improvements result in an exploration strategy that can learn a robust generalist policy.

Still, we find such an exploration strategy above is not enough to learn specialist policies from scratch on the terrains in Fig. 1. Besides, we want the learned sensorimotor skills to be reusable for new terrains. To this end, we divide the training procedure into two stages, where the first stage is to learn a common generalist policy on sparse and disorderly stepping stones, and the second stage is to finetune different specialist policies on different terrains from this generalist policy. By doing so, the generalist policy can learn some basic motion skills that are reusable on different terrains, and the specialist policies can benefit from the pre-training of the first stage. This generalist-specialist training procedure enables learning of agile locomotion skills which are then transferred to a real robot on risky terrains.

It is also worth mentioning that we do not focus on perception in this paper and therefore use a ground truth map and a motion capture system for state estimation during real-world deployment. That said, it is possible to use onboard sensors with the techniques proposed in [23].

In summary, our contributions include:

1) The extension of the navigation formulation in [20] to challenging risky terrains;
2) An exploration strategy consisting of a curriculum, intrinsic rewards, and a symmetry-based data augmentation;
3) The methodology of training a generalist policy and finetuning specialist policies under hard exploration;
4) Successful sim-to-real transfer of the learned agile motions on challenging stepping stones and balance beams.

II. Methodology

All of the implementations in this paper are based on the PPO algorithm [24] with massively parallel simulation [19] enabled by the GPU-based simulator Isaac Gym [25]. An overview is presented in Fig. 2.

A. Formulating Locomotion as Navigation

In this paper, we formulate the problem as a navigation task: the robot should get close to an assigned target on the terrain to get high task rewards. The navigation formulation has shown great potential when velocity commands cannot be easily assigned, e.g., when the robot walks in a maze [26] or needs to first recover from a fall and then walk [27]. In [20], advanced locomotion skills like jumping over large gaps and climbing high boxes are learned with the navigation formulation, where velocity-based tracking fails to achieve the desired behaviors.

In this work, considering that the robot should adapt its velocity to the terrain and inappropriate velocity commands can be misguiding, we also use the navigation formulation. We deploy the system illustrated in Fig. 2 based on [20], and assign position and heading commands to the robots. Tracking these commands dominates the rewards as follows:

$$r_{\text{task}} = \begin{cases} \frac{c_{\text{task}}}{T_r} \cdot \frac{1}{1+\|\chi_t - \chi_t^*\|^2}, & \text{if } t > T - T_r \\ 0, & \text{otherwise,} \end{cases} \quad (1)$$

where χ and χ^* are the current and target positions (or headings) of the base, t is the current time, T is the maximum episode length, and T_r defines the duration of the reward. By doing so, the policy is free to modulate the movement of the robot as long as it reaches the target towards the end of the episode.

B. Exploration Strategy

Our exploration strategy improves the prior work [20] in three aspects: the curriculum learning design, the intrinsic

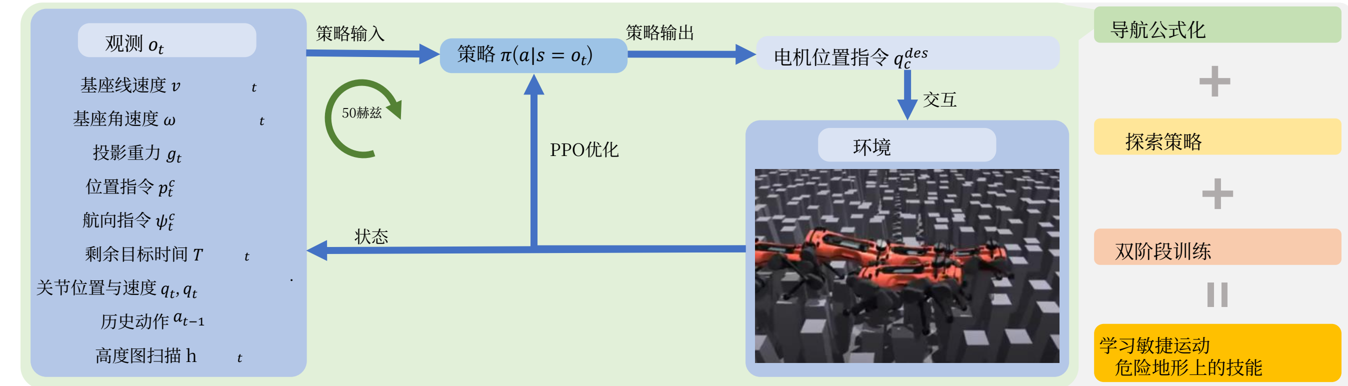

图2. 我们方法论的总览。左侧展示了学习系统。右侧列出了不同的组成部分。

基于常用的速度跟踪公式化（例如 [19] 和[12]中使用的）的传统学习方法无法完成这些具有挑战性的任务。在危险地形上，尤其是当机器人需要快速调整速度时，为其分配合适的速度指令是难以处理的，而且与非危险地形相比，可行解非常稀少且难以探索。因此，本文的关键思想是像 [20], 中那样将任务表述为一个导航问题，尽管危险环境和稀疏奖励使得探索更加困难。

为了克服导航公式中的困难探索问题，我们改进了 [20],中的课程学习设计，并融入了好奇心驱动的内在奖励 [21]。我们还采用了基于 [22],的基于对称性的数据增强，利用我们机器人的左右和前后对称性来提高数据效率。这些改进形成了一种能够学习鲁棒通用策略的探索策略。

尽管如此，我们发现上述探索策略仍不足以在图1中的地形上从头训练专家策略。此外，我们希望学习到的感觉运动技能能够在新地形上复用。为此，我们将训练流程分为两个阶段，第一阶段是在稀疏且无序的踏脚石上学习一个通用的通用策略，第二阶段是从该通用策略出发，在不同地形上微调不同的专家策略。通过这种方式，通用策略可以学习到一些在不同地形上可复用的基本运动技能，而专家策略则可以受益于第一阶段的预训练。这种通用-专家训练流程使得学习敏捷运动技能成为可能，并随后将这些技能迁移到危险地形上的真实机器人。

还值得一提的是，本文不关注感知问题，因此在真实世界部署中使用真值地图和动作捕捉系统进行状态估计。尽管如此，也可以结合 [23]中提出的技术使用机载传感器。

综上所述，我们的贡献包括：

1) 将导航公式化扩展到 [20]具有挑战性的危险地形；
2) 一种由课程、内在奖励和基于对称性的数据增强组成的探索策略；

3) 在困难探索条件下训练通用策略并微调专家策略的方法论；4) 在具有挑战性的踏脚石和平衡木上成功实现学习到的敏捷动作的仿真到现实迁移。

II. 方法论

本文中的所有实现均基于PPO算法 [24]，并借助基于GPU的仿真器Isaac Gym [25]实现大规模并行仿真 [19]。整体概览如图2所示。

A. 将运动控制公式化为导航任务

本文将问题公式化为导航任务：机器人应接近地形上指定的目标以获得高任务奖励。当速度指令难以直接指定时，例如机器人在迷宫[26] 中行走或需要先从跌倒中恢复再行走 [27]时，导航公式化展现出巨大潜力。在 [20], 中，跳跃大间隙和攀爬高台等高级运动技能通过导航公式化得以学习，而基于速度的跟踪无法实现这些期望行为。

在本工作中，考虑到机器人应适应地形调整自身速度，且不恰当的速度指令可能产生误导，我们也采用了导航公式化。我们基于 [20],部署了图2所示的系统，并向机器人分配位置和航向指令。对这些指令的跟踪主导了奖励，具体如下：

$$r_{\text{task}} = \begin{cases} \frac{c_{\text{task}}}{T_r} \cdot \frac{1}{1+\|\chi_t - \chi_t^*\|^2}, & \text{if } t > T - T_r \\ 0, & \text{otherwise,} \end{cases} \quad (1)$$

其中 χ 和 χ^* 分别是基座的当前和目标位置（或航向），t 是当前时间，T 是最大回合长度，T_r 定义了奖励的持续时间。通过这种方式，只要机器人在回合接近结束时到达目标，策略就可以自由调节机器人的运动。

B. 探索策略

我们的探索策略在三个方面改进了先前工作 [20]：课程学习设计、内在

rewards, and the data augmentation. The proposed techniques can empirically train a more robust policy under sparse rewards, and are experimentally analyzed in the ablation studies in Sec. IV-B.

1) Curriculum Design: Curriculum learning is a common technique for learning locomotion skills on challenging terrains [7], [10], [19], [20]. In Isaac Gym, a typical implementation of curriculum learning is to generate all terrains of different difficulty levels before training, and distribute the robots on different terrains [19], [20]. Based on this, the curriculum design in [20] is formulated as: 1) go to a higher difficulty level if the policy performs well, 2) go to a lower difficulty level if the policy performs poorly, and 3) go to a random difficulty level if the policy performs well on the highest difficulty level.

However, we find that such a curriculum suffers from sparse rewards and difficulty gaps between levels due to the highly challenging and risky terrains. A policy performing well on a low-difficulty terrain can hardly work on a high-difficulty terrain, leading to a large reward difference that can be incorrectly attributed by the difficulty-agnostic optimizer to different actions, thereby misguiding the optimization. To alleviate this problem, we propose to relax the curriculum update mechanism: the robot only gets demoted when the policy cannot outperform random actions in the remaining distance to the target. By doing so, the robot can keep interacting with the same level until it overcomes it.

2) Intrinsic Curiosity Rewards: On top of the task rewards which incentivize the robot to reach the target, we also use curiosity-based intrinsic rewards that encourage the robot to explore novel state-action pairs during training. Among many variants [21], [28], [29], we use the random network distillation (RND) [21] with the version in [30] due to its simple implementation and competitive performance against other variants. Specifically, the curiosity reward is defined as

$$r_{\text{curio}}(s,a) = c_{\text{curio}} \cdot |M_1(s,a) - M_2(s,a)|, \tag{2}$$

where M_1 and M_2 are two randomly initialized neural networks taking state-action (s,a) pairs as inputs, and M_1 can be trained while M_2 is frozen. By fitting M_1 towards the outputs of M_2 with visited state-action pairs, the curiosity rewards will be high for unfamiliar pairs but decay to 0 for familiar ones.

It is worth mentioning that while curiosity rewards are commonly leveraged to improve data efficiency, we do not witness a significant change on the learning curves. Instead, we empirically find that it can improve the robustness of the learned policy, as is discussed in Sec. IV-B. This can potentially be due to an increased coverage of the state space, as suggested by [31] where a larger state-space coverage can improve the robustness of some learned locomotion skills.

3) Symmetry-Based Data Augmentation: To improve the data efficiency and reduce base inclination during locomotion, we utilize the symmetric design of our robot to augment sampled data as proposed in [22], which is an adaptation of the duplication method in [32]. To be specific, one state-action pair can be augmented to four pairs with left-right and front-back symmetry. The augmented state-action pairs are assigned the same target values, advantages, and discounted returns as the original pairs. In this way, the actor and critic networks are enforced to learn symmetric actions and values.

C. Double-Stage Generalist-Specialist Training

In this paper, we achieve agile locomotion on diverse challenging terrains with sparse footholds as shown in Fig. 1. However, even with all the techniques presented above, these motions cannot be learned from scratch due to the extremely sparse rewards. To address this, we first train a generalist policy on relatively simple stepping stones (see Sec. III-A). Then, the learned policy and its critic network are finetuned on new terrains, enabling data-efficient learning of agile motions, where the generalist policy can serve as a pre-trained model.

Besides the data efficiency, the generalist policy is shared for different specialist policies, which means that the learned sensorimotor skills are reusable. Therefore, the costly generalist training stage only happens once, and only data-efficient finetuning is needed for new terrains.

III. Experimental Setup

We create 4 different types of terrains in Isaac Gym, each with 10 difficulty levels. Among them, "Stones-Everywhere" is used for generalist policy training and benchmarking, while the other three ("Stones-2Rows", "Balance-Beams", and "Stepping-Beams") are used for finetuning. The terrains are described in Sec. III-A. Sec. III-B and Sec. III-C present the specific rewards and task-related randomizations of each terrain type respectively.

A. Terrains

The four terrain types are displayed in Fig. 3 and described in the following paragraphs. Since the "Stones-Everywhere" type serves to benchmark the performance of different learning methods, we present it in detail, while other terrain types are introduced briefly.

1) Stones-Everywhere: This is the terrain type used to train the generalist policy and benchmark the ablation studies. Each stone is assigned a grid square and has a random height with a random X-Y plane shift from the square center. Highly disorderly and sparse stepping stones can be generated in this way. The parameters for different levels include the stone size w_{stone}, the stone gap w_{gap}, the maximum shift $s_{\max}$, and the maximum height $h_{\max}$. The stone size is the width of the square stones, and the stone gap is the grid size minus the stone size, i.e., the size of gap between two stones with zero shift. Both the X shift and the Y shift are uniformly sampled from [-$s_{\max}$, $s_{\max}$], and the stone height is uniformly sampled from [-$h_{\max}$, $h_{\max}$].

The parameters are presented in Table I, together with the sparsity representing the ratio of the non-steppable area over the whole terrain.

2) Stones-2Rows: Terrains of this type have 1, 2 or 3 rows of stepping stones which match what we can build in the lab. By doing so we can achieve a policy that does not overfit to a specific gait pattern.

奖励以及数据增强。所提出的方法能够在稀疏奖励下经验性地训练出更稳健的策略，并在第IV-B节的消融研究中进行了实验分析。

1) 课程设计：课程学习是在具有挑战性的危险地形上学习运动技能的常用技术 [7], [10], [19], [20]。在Isaac Gym 中，课程学习的典型实现方式是在训练前生成所有不同难度等级的地形，并将机器人分布在不同地形上 [19], [20]。基于此， [20] 中的课程设计被表述为：1) 如果策略表现良好，则进入更高难度等级；2) 如果策略表现不佳，则进入更低难度等级；3) 如果策略在最高难度等级上表现良好，则进入随机难度等级。

然而，我们发现这种课程由于地形极具挑战性和危险性，存在稀疏奖励和等级间难度差距的问题。在低难度地形上表现良好的策略很难在更高难度的地形上发挥作用，导致较大的奖励差异，而难度无关优化器可能会错误地将这种差异归因于不同的动作，从而误导优化过程。为了缓解这一问题，我们提出放宽课程更新机制：只有当策略在到目标的剩余距离上无法优于随机动作时，机器人才会被降级。通过这种方式，机器人可以持续与同一等级交互，直到克服该等级。

2) 内在好奇心奖励：在激励机器人到达目标的任务奖励之上，我们还使用基于好奇心的内在奖励，以鼓励机器人在训练过程中探索新颖的状态-动作对。在许多变体中， [21], [28], [29], 我们采用随机网络蒸馏（RND） [21] ，其版本来自 [30]，因为它实现简单且性能优于其他变体。具体而言，好奇心奖励定义为

$$r_{\text{curio}}(s,a) = c_{\text{curio}} \cdot |M_1(s,a) - M_2(s,a)|, \tag{2}$$

其中M_1 和M_2 是两个随机初始化的神经网络，以状态-动作（s,a）对作为输入，M_1 可被训练，而M_2 保持冻结。通过使用已访问的状态-动作对将M_1 拟合至M_2 的输出，对于不熟悉的状态-动作对，好奇心奖励会很高，而对于熟悉的状态-动作对则会衰减至0。

值得一提的是，虽然好奇心奖励通常被用来提升数据效率，但我们并未观察到学习曲线的显著变化。相反，我们通过实验发现它可以提高所学策略的鲁棒性，如第IV-B节所讨论。这可能归因于状态空间覆盖范围的扩大，正如 [31]所建议的，更大的状态空间覆盖可以提高某些学习到的运动技能的鲁棒性。

3) 基于对称性的数据增强：为了提高数据效率并减少运动过程中的基底倾斜，我们利用机器人的对称设计来增强采样数据，如 [22],中所提出的方法，该方法是对 [32]中复制方法的改编。具体而言，一个状态-动作对可以通过左右和前后对称性增强为四个状态-动作对。增强状态-动作对被赋予与原始状态-动作对相同的目标值、优势函数和折扣回报。通过这种方式，演员与评论家网络被强制学习对称的动作和值。

C. 两阶段通才-专才训练

在本文中，我们实现了在具有稀疏立足点的多样挑战性地形上的敏捷运动，如图1所示。然而，即使采用了上述所有技术，由于奖励极度稀疏，这些动作也无法从头训练。为解决这一问题，我们首先在相对简单的踏脚石上训练一个通用策略（见第III-A节）。然后，将学习到的策略及其评论家网络在新地形上进行微调，从而实现敏捷动作的数据高效学习，其中通用策略可作为预训练模型。

除了数据效率之外，通用策略还被不同的专家策略共享，这意味着学习到的感觉运动技能是可复用的。因此，成本高昂的通用训练阶段仅需进行一次，对于新地形只需进行数据高效的微调即可。

III. 实验设置

我们在Isaac Gym中创建了4种不同类型的地形，每种地形有10个难度级别。其中，“Stones-Everywhere”用于通用策略训练和基准测试，而其他三种（“Stones-2Rows”、“Balance-Beams”和“Stepping-Beams”）用于微调。这些地形在第III-A节中描述。第III-B节和第III-C节分别介绍了每种地形类型的特定奖励和与任务相关的随机化设置。

A. 地形

四种地形类型如图3所示，并在以下段落中加以描述。由于“Stones-Everywhere”类型用于基准测试不同学习方法的性能，我们将对其进行详细介绍，而其他地形类型则简要介绍。

1) Stones-Everywhere：这是用于训练通用策略并基准测试消融研究的地形类型。每块石头被分配一个网格方块，并具有从方块中心随机偏移的X-Y平面偏移的随机高度。通过这种方式可以生成高度无序且稀疏的踏脚石。不同级别的参数包括石头尺寸w_{stone}、石头间隙w_{gap}、最大偏移$s_{\max}$和最大高度$h_{\max}$。石头尺寸是方形石头的宽度，石头间隙是网格尺寸减去石头尺寸，即两块零偏移石头之间的间隙大小。X偏移和Y偏移均从 [-$s_{\max}$、$s_{\max}$]中均匀采样，石头高度从 [-$h_{\max}$、$h_{\max}$]中均匀采样。

参数见表I，其中稀疏度表示不可踏区域占整个地形的比例。

2) Stones-2Rows：此类地形包含1、2或3行踏脚石，与我们在实验室中可搭建的结构相匹配。通过这种方式，我们可以获得一个不会过度拟合特定步态模式的策略。

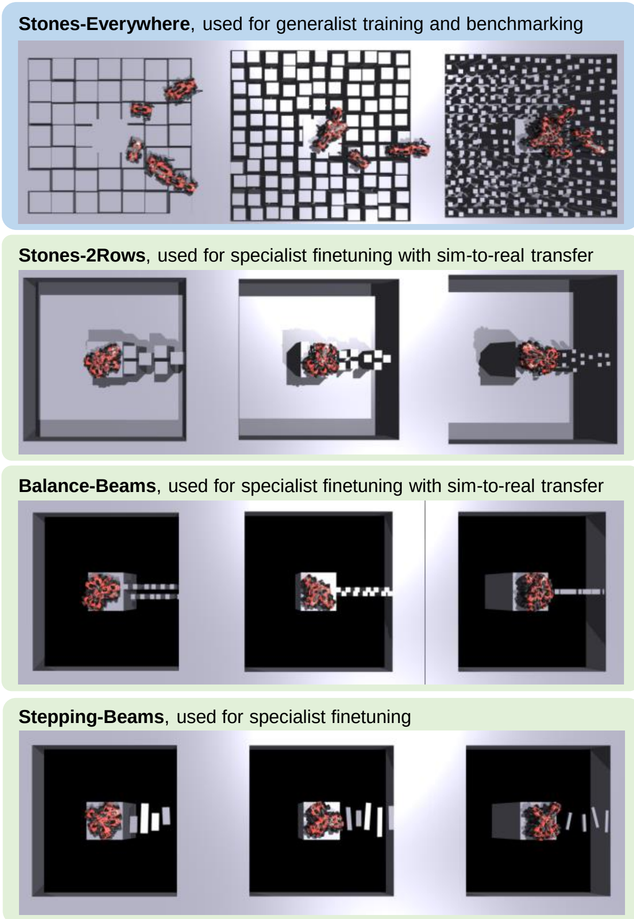

Fig. 3. Different types of terrains with difficulty levels 0, 5, and 9 from left to right. One terrain type can have different subtypes that stress different features, as is explained in Sec. III-A.

3) Balance-Beams: Terrains of this type have stepping stones evolving from two rows merging into one as the level goes up. Random gaps are added to the terrains as well. Such evolving stones can guide the policy to learn catwalk-like motions. Beams with gradually smaller widths can be a more intuitive choice, but this can lead to infeasible crawling motions in Fig. 4, a local minimum that works only on wide beams.

4) Stepping-Beams: Terrains of this type have a sequence of beams to step on, and the beams are more narrow and randomized on higher levels. This terrain type is hard to build in the lab so we only tried it in simulation to show that finetuning can work on a wide range of different terrains.

B. Rewards

The same reward function can work on all terrains in simulation, yet slight terrain-dependant modifications of are necessary for elegant motions in the real world. We present the rewards in Table IV in the Appendix, and the changes in weights for special tasks are annotated. For each task on its terrain, the reward function is the sum of all terms.

Among these rewards, we have position tracking and heading tracking for task performance. We use penalties for early termination and collision. We also regularize the joint velocities, the acceleration of the base and feet, the action rate, the torques, and the contact forces. We guide exploration with the "Don't wait" and "Move in Direction" rewards as in [20], and provide a "Stand Still" rewards to make the robot stand still after reaching the target.

We regularize the aggressive motions on "Stones-2Rows" and "Balance-Beams", and further regularize the torques on "Balance-Beams". We also add a "Stand Pose" reward to "Stones-2Rows" and "Balance-Beams" to prepare the robot for further repetitive tests.

TABLE I

PARAMETERS OF "STONES-EVERYWHERE" TERRAINS

Difficulty Level	Stone Size w_{stone} (cm)	Stone Gap w_{gap} (cm)	Max. Shift $s_{\max}$ (cm)	Max. Height $h_{\max}$ (cm)	Sparsity (%)
0	92	8	3.6	1	15.4
…	…	…	…	…	
5	52	18	8.2	6	44.8
…	…	…	…	…	
9	20	26	11.8	10	81.1

* All parameters except the calculated sparsity are linear to the levels so we only list the levels 0, 5, and 9.

** For reference, [12] built the stepping stones of $\sim 65\%$ sparsity without random heights and significant shifts.

Fig. 4. The learned infeasible crawling motions on balance beams due to inappropriate terrain curriculum design. The terrain here for display is far less challenging than the terrains in the real world, but the robot cannot traverse it. When the beam is narrow, the robot cannot find its next step on the beam.

C. Domain Randomization

We use domain randomization to improve robustness and facilitate sim-to-real transfer [33], [34]. Task-related randomization settings are presented in Table III in the Appendix. We slightly extend the episode length to slow down the aggressive motions on "Stones-2Rows" and "Balance-Beams". The center of mass (CoM) is heavily randomized for "Balance-Beams" to reduce the dangerous shaking of the base on the beams.

IV. RESULTS IN SIMULATION

A. Learned Motions in Simulation

Our learned policy demonstrates many interesting behaviors on "Stones-Everywhere" in simulation. In Fig. 5, we made the robot walk around on stepping stones by providing a series of waypoints as target positions and headings, showcasing the ability of omnidirectional locomotion learned by our method. During locomotion, the robot also had different gait patterns and contact modes under different situations.

We also find the policy can recover from missed steps in simulation, as is displayed in Fig. 6. The ~ 50 kg robot is able to quickly react to the external force disturbances and continues to walk even when the forces on all feet reaches ~ 100 N.

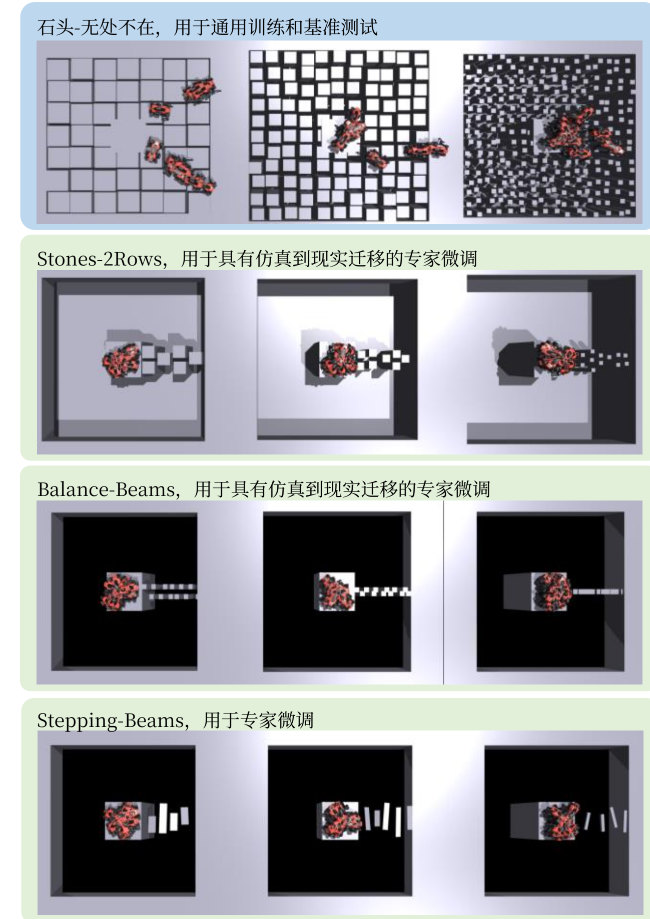

图3. 从左到右依次为难度等级0、5和9的不同类型地形。一种地形类型可以有强调不同特征的不同子类型，如第III-A节所述。

3) Balance-Beams：此类地形的踏脚石随着难度等级提升，从两行逐渐合并为一行。地形中还会随机添加间隙。这种逐渐演变的石头可以引导策略学习类似猫步的运动。逐渐变窄的横梁可能是一个更直观的选择，但这可能导致图4中不可行的爬行运动，即一种仅适用于宽横梁的局部最小值。

4) Stepping-Beams：此类地形具有一系列可供踩踏的横梁，在更高等级中横梁更窄且更随机。这种地形类型难以在实验室中搭建，因此我们仅在仿真中尝试了它，以表明微调可以适用于各种不同的地形。

B. 奖励

相同的奖励函数可以在仿真中的所有地形上工作，但为了实现真实世界中的优雅动作，需要根据地形进行细微的调整。我们在附录的表IV中展示了奖励项，并标注了特殊任务中权重的变化。对于每种地形上的每个任务，奖励函数是所有项的总和。

在这些奖励中，我们使用位置跟踪和航向跟踪来衡量任务表现。我们使用惩罚项来处理提前终止和碰撞。我们还对关节速度、基座加速度、脚部加速度以及动作进行正则化处理，包括动作变化率、扭矩和接触力。我们按照 [20], 中的方法，使用“不要等待”和“按方向移动”奖励来引导探索，并提供“保持静止”奖励，使机器人在到达目标后保持静止。

我们对“Stones-2Rows”和“Balance-Beams”上的激进运动进行正则化，并进一步对“Balance-Beams”上的扭矩进行正则化。我们还在“Stones-2Rows”和“Balance-Beams”上添加了“站立姿态”奖励，以便为机器人进一步的重复测试做好准备。

表I “STONES-EVERYWHERE”地形的参数

难度等级	石头尺寸 宽$_{\text{stone}}$ (厘米)	石头间隙 宽$_{\text{gap}}$ (厘米)	最大偏移 间距$_{\max}$ (厘米)	最大高度 高$_{\max}$ (厘米)	稀疏度 (%)
0	92	8	3.6	1	15.4
…	…	…	…	…	
5	52	18	8.2	6	44.8
…	…	…	…	…	
9	20	26	11.8	10	81.1

*除计算出的稀疏度外，所有参数均随等级线性变化，因此我们仅列出等级0、5和9。

** 作为参考， [12] 构建了 $\sim 65\%$稀疏度的踏脚石，未设置随机高度和显著偏移。

图4. 由于不恰当的地形课程设计，机器人在平衡木上学习到了不可行的爬行运动。此处展示的地形远不如真实世界中的地形具有挑战性，但机器人仍无法穿越。当横梁较窄时，机器人无法在横梁上找到下一步的落点。

C. 域随机化

我们使用域随机化来提高鲁棒性并促进仿真到现实迁移 [33], [34]。与任务相关的随机化设置见附录中的表III。我们略微延长了回合长度，以减缓“Stones-2Rows”和“Balance-Beams”上的激进运动。对于“Balance-Beams”，质心被大幅随机化，以减少基座在横梁上的危险晃动。

IV. 仿真结果

A. 仿真中学习到的运动

我们学习到的策略在仿真中的“Stones-Everywhere”上展现了许多有趣的行为。在图5中，我们通过提供一系列航路点作为目标位置和航向，使机器人在踏脚石上行走，展示了我们的方法学习到的全向运动能力。在运动过程中，机器人还在不同情况下表现出不同的步态模式和接触模式。

我们还发现，该策略能够在仿真中从失足中恢复，如图6所示。 ~ 50 公斤的机器人能够快速应对外部力干扰，即使所有脚上的力达到~ 100 牛，也能继续行走。

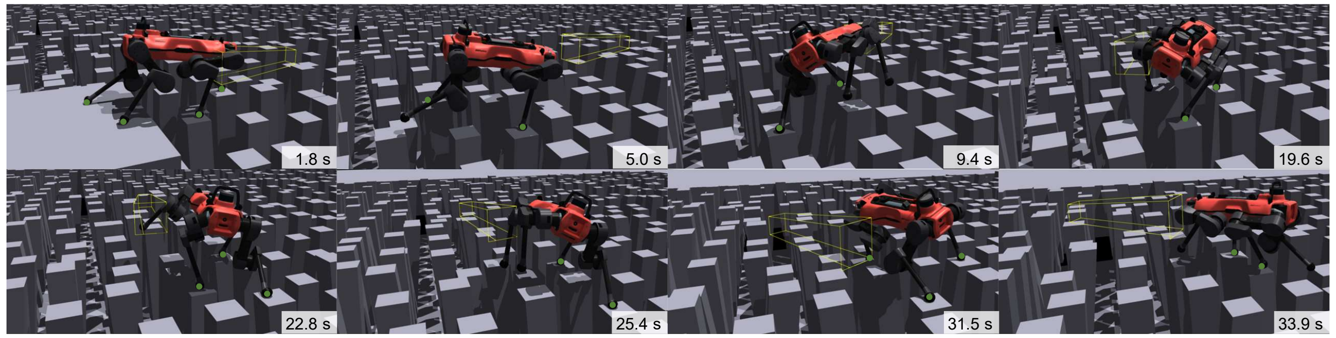

Fig. 5. The robot walking around on the hardest "Stones-Everywhere" terrain in simulation. The yellow boxes indicate a series of target positions and headings we provided to the robot to achieve omnidirectional locomotion. The robot had different gait patterns under different situations, e.g., the 3-contact walk or 2-contact trot gaits when easy to find safe footholds ahead, and the 2-contact pace gaits otherwise. Feet in contact with the terrain are marked.

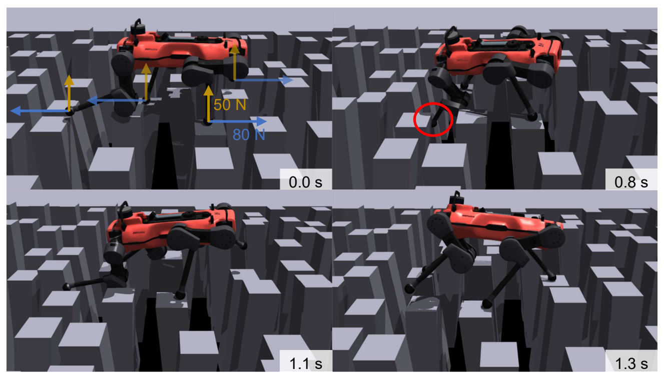

Fig. 6. The robot could dynamically recover from missed steps due to external force disturbances while walking on stepping stones in simulation. **Left top:** the robot received external forces on its four feet lasting 0.4 s when traversing the stones from left to right. **Right top:** the robot missed a step due to the external forces. **Left bottom:** the robot recovered from the missed step by pulling itself upwards and forwards with the legs pressing the stones. **Right bottom:** the robot continued to walk after the missed step.

B. Benchmarked Ablation Studies

We compare different settings and analyze the robustness of the learned policies on "Stones-Everywhere". The compared settings are:

1) w/o navigation: where the velocity tracking formulation is used based on [19] with the corresponding curriculum, while the intrinsic rewards and the symmetry-based data augmentation are also applied;
2) w/o curriculum: every technique proposed in this paper except the improved curriculum is used, and the curriculum is the one in [20];
3) w/o curiosity: every technique proposed in this paper except the intrinsic curiosity rewards is used;
4) w/o augmentation: every technique proposed in this paper except the data augmentation is used;
5) proposed: every technique proposed in this paper is used.

Two metrics are used for the comparison of performance and robustness. The first one is the success rate of traversing 6 meters on different difficulty levels, as is shown in Fig. 7. After level 4 with sparsity $\sim 40\%$, the averaged success rate for velocity tracking is lower than 90% and quickly drops down as the difficulty level goes up. In comparison,

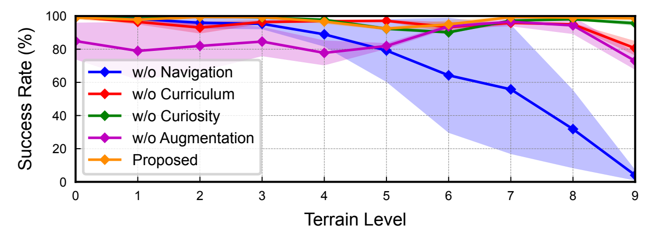

Fig. 7. Success rate of traversing 6 meters on different levels of "Stones-Everywhere". The settings based on the navigation formulation far outperform the velocity tracking one. Three runs (policies trained with different random seeds) per setting were performed to calculate the mean (line) and std (in shadow) values, and the checkpoints of 8000 iterations (all converged) were used. The velocity command used for testing velocity tracking is 0.8 m/s forward, which demonstrates the highest robustness empirically in simulation. Each policy's success rate is calculated from 1000 trials on 100 diverse randomized terrains.

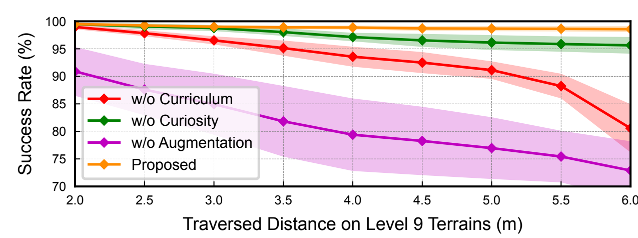

Fig. 8. Success rate of traversing different distances on the hardest level of "Stones-Everywhere". Our proposed strategy stably learns robust policies over long distances. The shadow indicates the std values.

the navigation formulation can achieve a high success rate, though the "w/o augmentation" setting does not perform well on low-difficulty terrains. This can be explained by the policy's inability to overcome the hardest terrains without data augmentation, and the neural network policy forgets the samples on low-difficulty terrains when the robots get stuck on high-difficulty ones.

The second metric is the success rate of traversing different distances on the highest difficulty level, which demonstrates a clearer trend between the navigation-formulated ones, as is shown in Fig. 8. All techniques in our strategy significantly contribute to improving robustness, with symmetry-based data augmentation being the most critical one. The improved curriculum design and the curiosity rewards have relatively smaller effects. Besides, the symmetry-based data augmentation also makes the motions more balanced with less base inclination.

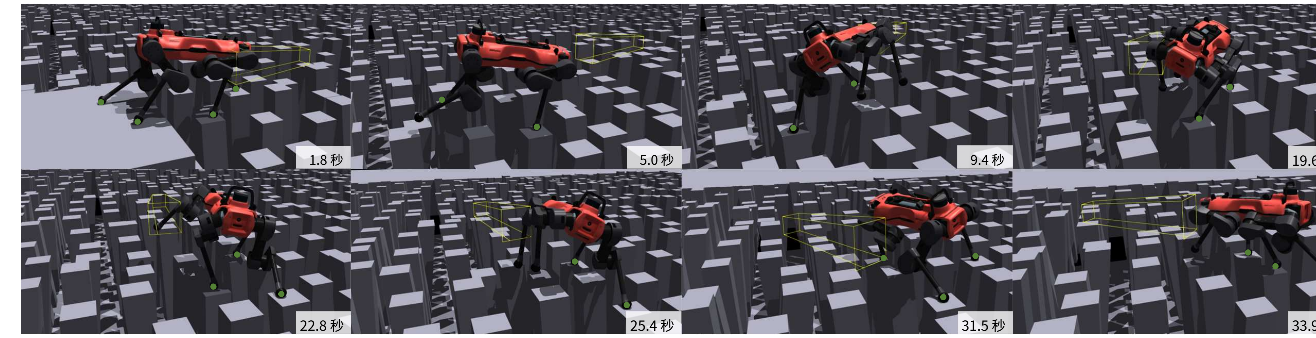

图5. 机器人在仿真中最困难的“Stones-Everywhere”地形上行走。黄色方框表示我们提供给机器人以实现全方位运动的一系列目标位置和航向。机器人在不同情况下采用不同的步态模式，例如，当前方容易找到安全落脚点时采用三足接触行走或两足接触快步态，否则采用两足接触踱步步态。与地形接触的脚已标记。

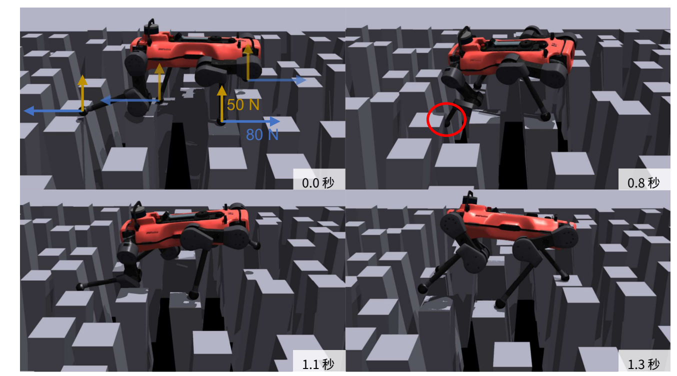

图6. 在仿真中行走于踏脚石上时，机器人能够从外力干扰导致的踏空步态中动态恢复。**左上：**机器人在从左向右穿越石头时，其四只脚受到持续0.4秒的外力。**右上：**机器人因外力而踏空一步。**左下：**机器人通过用腿按压石头将自身向上和向前拉动，从踏空步态中恢复。**右下：**机器人在踏空后继续行走。

B. 基准消融研究

我们比较了不同设置，并分析了学习到的策略在“Stones-Everywhere”上的鲁棒性。比较的设置包括：

1) 无导航：使用基于 [19] 的速度跟踪公式化，并配合相应的课程，同时应用内在奖励和基于对称性的数据增强；2) 无课程：使用本文提出的除改进课程外的所有技术，课程采用 [20]中的方案；3) 无好奇心：使用本文提出的除内在好奇心奖励外的所有技术；4) 无增强：使用本文提出的除数据增强外的所有技术；5) 所提出的方法：使用本文提出的所有技术。

我们使用两个指标来比较性能和鲁棒性。第一个指标是在不同难度等级下穿越6米距离的成功率，如图7所示。在稀疏度为 $\sim 40\%$的第4级之后，速度跟踪的平均成功率低于90%，并随着难度等级的升高而迅速下降。相比之下，

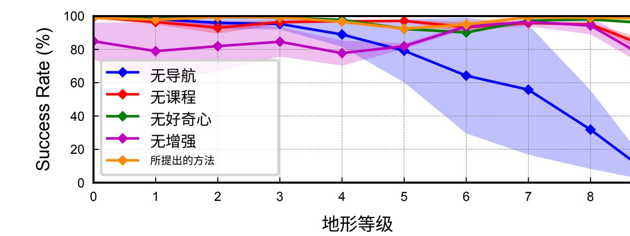

图7. 在不同等级的“Stones-Everywhere”上穿越6米的成功率。基于导航公式化的设置在性能上远超速度跟踪设置。每种设置进行了三次运行（使用不同随机种子训练的策略），以计算平均值（线）和标准差（阴影区域）的值，以及8000次迭代的检查点。
（全部收敛）被使用。用于测试速度跟踪的速度指令为0.8米/秒前进，这在仿真中经验性地表现出最高的鲁棒性。每个策略的成功率是根据100个多样化的随机地形上的1000次试验计算得出的。

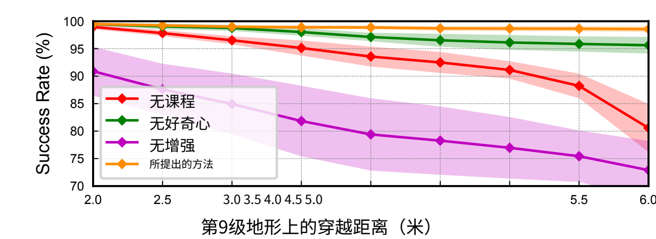

图8. 在“Stones-Everywhere”最难级别上穿越不同距离的成功率。我们所提出的方法能够在长距离上稳定地学习鲁棒策略。阴影区域表示标准差的值。

导航公式化能够实现较高的成功率，尽管“无增强”设置在地形难度较低时表现不佳。这可以解释为：没有数据增强，策略无法克服最困难的地形，而当机器人卡在困难地形上时，神经网络策略会遗忘低难度地形上的样本。

第二个指标是在最高难度等级下穿越不同距离的成功率，如图8所示，该指标在基于导航公式的方法之间呈现出更清晰的趋势。我们策略中的所有技术都对提升鲁棒性有显著贡献，其中基于对称性的数据增强最为关键。改进后的课程设计和好奇心奖励的影响相对较小。此外，基于对称性的数据增强还使运动更加平衡，基底倾斜更小。

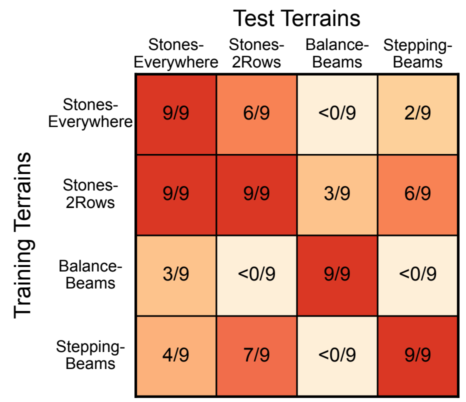

Fig. 9. The transferability matrix of our learned policies. Values (before "/9") in the matrix are the maximum difficulty ($0 \sim 9$ with 9 being the hardest) of test terrains that can be traversed for 3.5m with an 80% success rate. For the "< 0"s, the success rate is within $[50\%, 80\%)$ on level 0.

C. Finetuning v.s. From Scratch

We train specialist policies on multiple challenging terrains as shown in Fig. 1 and 3 by finetuning the learned generalist policy. The generalist policy took 8000 iterations, the "Stones-2Rows" policy took 2000 iterations, the "Balance-Beams" policy took 3000 iterations, and the "Stepping-Beams" policy took 1000 iterations. In comparison, when the specialist policies are trained from scratch for 10000 iterations per task, the robot can not learn to traverse the terrains, as we design these terrains to be quite challenging.

D. Transferability of Policies

We also test whether policies trained on different terrains are transferable to other terrains. Figure 9 presents the transferability matrix of our learned policies. Due to the similarity of terrains, policies trained on "Stones-Everywhere" and "Stones-2Rows" can be mutually transferred, and the latter is also transferrable to and from "Stepping-Beams". For other terrain pairs, the policies do not tranfer well. Such results indicate that the learned policies highly differentiate due to their training data although they originate from the same generalist policy. Hence, if we want a unified policy that performs well on various terrains, we must randomize all possible terrains the robot may encounter. Otherwise, the learned policy cannot generalize well.

V. Real-World Results

Due to practical reasons, we only build two terrains in the real world: stepping stones and balance beams, and successfully transfer the learned "Stones-2Rows" and "Balance-Beams" policies. To minimize the sim-to-real gap and focus on the locomotion problem, we use a motion capture system to get the robot's state and terrain information.

A. Agile Motions

Motions of the learned policies on real terrains are displayed in Fig. 10 and 11. The motions involve multiple leaps and rapidly changing velocities which are difficult to achieve with velocity tracking. To solve the task in the given episode length, the robot takes different strategies on

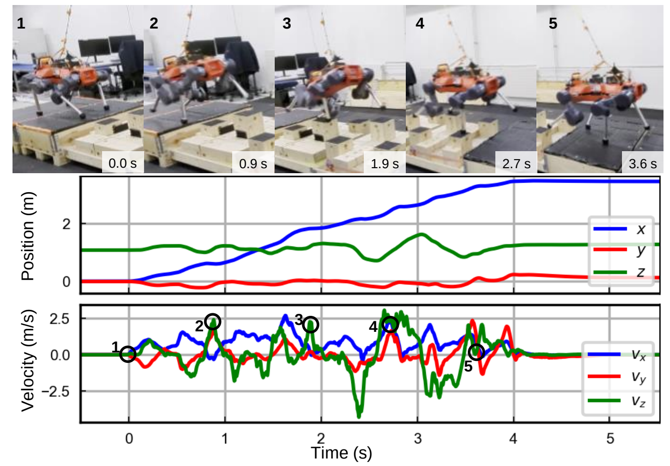

Fig. 10. Learned agile motions on stepping stones in real-world experiments. The robot frequently leaps upwards and forwards to traverse the terrain. The robot positions are sampled from the motion capture system in the world frame, and the velocities are obtained by differentiating the positions.

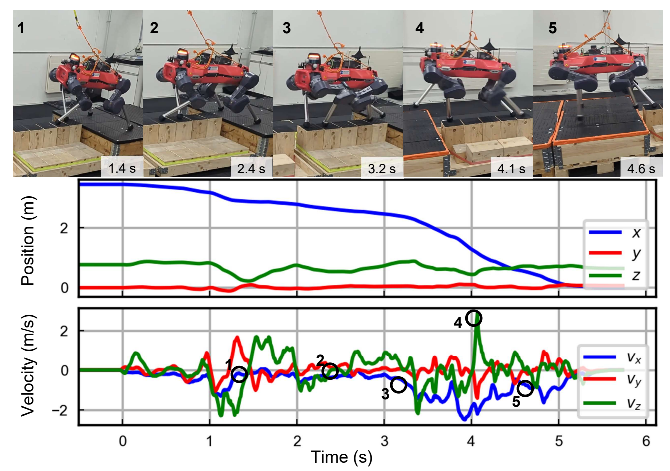

Fig. 11. Learned agile motions on balance beams in real-world experiments. The robot first slowly aligns itself with the beam center before landing on the beam. Afterward, the robot rapidly alternates its left and right diagonals, and makes a speed burst ($3 \sim 5$ s in the plot) to traverse the beams.

TABLE II
Repetitive Robustness Test Results on Real Terrains

Real Terrain	Stones-2Rows	Balance-Beams
Success Rate	8/10	5/5
Survival Rate	10/10	5/5

different terrains. It keeps leaping on stepping stones, while first slowly aligning itself towards the center and then making a speed burst to quickly traverse the terrain on balance beams.

B. Robustness Tests

We did repetitive robustness tests for the two policies (see also the video attachment), and the results are presented in Table II. The success rate counts the trials that were perfectly finished, while the survival rate also counts the cases where the robot missed a step but recovered.

Our method also differs from the model-based controllers in that the robot can quickly react to unexpected missed steps and survive, as is discussed in Sec. IV-A. We refer readers

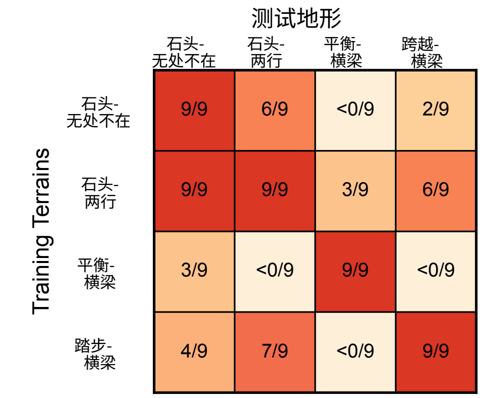

图 9. 我们学习到的策略的迁移性矩阵。矩阵中的值（“/9”之前）表示在测试地形上能够以80%的成功率穿越3.5米的最大难度（$0 \sim 9$ 其中9为最难）。对于“< 0”对应的值，在等级0上的成功率在 $[50\%, 80\%)$ 之间。

C. 微调与从头训练

我们通过对已学习的通用策略进行微调，在图1和图3所示的多个具有挑战性的地形上训练专家策略。通用策略训练了8000次迭代，"Stones-2Rows"策略训练了2000次迭代，"Balance-Beams"策略训练了3000次迭代，"Stepping-Beams"策略训练了1000次迭代。相比之下，当专家策略从零开始为每个任务训练10000次迭代时，机器人无法学会穿越这些地形，因为我们设计这些地形的难度相当高。

D. 策略的可迁移性

我们还测试了在不同地形上训练的策略是否可以迁移到其他地形。图9展示了我们学习到的策略的可迁移性矩阵。由于地形的相似性，在“Stones-Everywhere”和“Stones-2Rows”上训练的策略可以相互迁移，后者也可以与“Stepping-Beams”相互迁移。对于其他地形组合，策略迁移效果不佳。这些结果表明，尽管学习到的策略源自同一个通用策略，但由于训练数据的不同，它们之间产生了显著差异。因此，如果我们希望获得一个能在各种地形上表现良好的统一策略，就必须对机器人可能遇到的所有地形进行随机化。否则，学习到的策略无法很好地泛化。

V. 真实世界结果

出于实际原因，我们仅在真实世界中搭建了两种地形：踏脚石和平衡木，并成功迁移了学习到的“Stones-2Rows”和“Balance-Beams”策略。为了最小化仿真到现实的差距并专注于运动问题，我们使用动作捕捉系统获取机器人的状态和地形信息。

A. 敏捷动作

学习到的策略在真实地形上的运动如图10和图11所示。这些运动涉及多次跳跃和快速变化的速度，这些通过速度跟踪难以实现。为了在给定的回合长度内完成任务，机器人采取了不同的策略，

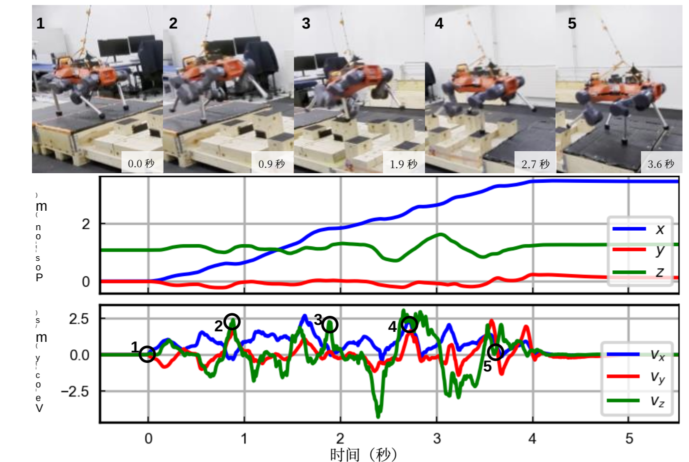

图 10. 在真实世界实验中，机器人在踏脚石上学习到的敏捷动作。机器人频繁地向上和向前跳跃，以穿越地形。机器人位置由世界坐标系中的动作捕捉系统采样获得，速度通过对位置求导得到。

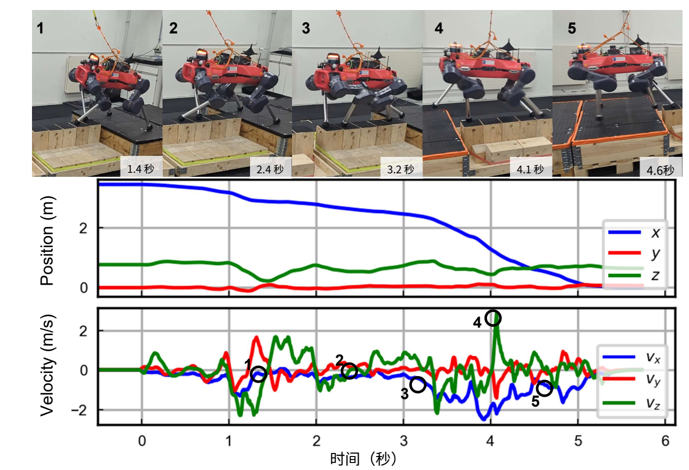

图11. 真实世界实验中平衡木上的学习型敏捷动作。机器人首先缓慢调整自身与横梁中心对齐，然后落在横梁上。随后，机器人快速交替切换左右对角线，并产生速度爆发（图中3～5 秒处）以穿越横梁。

表II 真实地形上的重复鲁棒性测试结果

真实地形	Stones-2Rows	Balance-Beams
成功率	8/10	5/5
存活率	10/10	5/5

针对不同的地形。在踏脚石上，它持续跳跃，而在平衡木上，它首先缓慢地将自身对准中心，然后进行速度爆发以快速穿越地形。

B. 鲁棒性测试

我们对这两种策略进行了重复鲁棒性测试（另见视频附件），结果如表II所示。成功率统计的是完美完成的试验次数，而存活率还统计了机器人踩空但成功恢复的情况。

我们的方法也与基于模型的控制器不同，机器人能够快速应对意外的踏空并存活下来，如第IV-A节所讨论。我们请读者

to the video attachment to see the emergent behaviors. It is worth mentioning that our controller is vulnerable to mapping drifts, and a 5-cm odometry drift can make the robot fall, though this is partly due to the highly risky terrains.

VI. Limitations and Discussion

Though we can achieve agile locomotion on various terrains, we have difficulty training a unified policy for all because the learning progresses on different terrains can vary and interfere with each other. Also, we use slightly different settings for different specialist tasks with human insights to help the sim-to-real transfer. The reward functions can be tedious to tune, and the rare failure cases are hard to interpret.

To be specific, the task reward functions are quite an "equilibrium" of multiple motivations. First, as we want the robot to automatically discover a way to reach the target without human priors, a short reward duration is preferred but can also lead to sparse rewards and hard exploration. Second, a long episode length can make the motions less aggressive for sim-to-real transfer but also makes the task rewards sparse. Hence, tuning the rewards is tedious, and we partly sidestep these trade–offs by pre-training and finetuning in this paper.

Regarding the failure cases, it is always difficult to achieve a 100% success rate, and the rare failure cases do exist. However, contrary to model-based controllers whose failure cases are easy to understand (e.g., missed steps, slippery, or unexpected collisions), our controllers' failure cases are more difficult to interpret. In these rare cases, the robot may misplace its feet in the air when there are feasible footholds, which might be due to the singularity of neural networks [35]. We believe representation learning techniques such as contrastive learning [36] may help solve this.

VII. Conclusion and Future Works

In this paper, we achieve end-to-end learning of agile quadrupedal locomotion on challenging risky terrains. To this end, we propose to use the navigation formulation and an exploration strategy with multiple improvements from previous works. We apply two-staged training to overcome exploration on challenging terrains, with specialist policies on different terrains that are finetuned from one generalist policy.

Our future work will include: 1) making the terrain perception onboard based on online representation techniques such as the one in [23]; 2) training a unified policy for locomotion on all kinds of terrains; and 3) making the learned policy more interpretable and robust.

APPENDIX

A. Task Specifications

The task-related domain randomization settings are presented in Table III, and the used symbols and reward terms in Table IV.

TABLE III
Domain Randomization Settings in Different Tasks

Parameter	Randomization Range
Stones-Everywhere	
Episode Length (s)	$[5,7]$
Target Distance (m)	$[1.5,4.9]$
Stones-2Rows	
Episode Length (s)	$[6,8]$
Target Distance (m)	$[3.5,4.5]$
Heightmap Scan Drift (m)	$[-0.05,0.05]$
Balance-Beams	
Episode Length (s)	$[8,10]$
Target Distance (m)	$[3.5,4.5]$
Heightmap Scan Drift (m)	$[-0.025,0.025]$
Center of Mass Bias (m)	$x: [-0.15,0.15]$ $y: [-0.05,0.05]$ $z: [-0.1,0.2]$
Stepping-Beams	
Episode Length (s)	$[5,7]$
Target Distance (m)	$[2.5,4.0]$

TABLE IV
Used Symbols and Reward Function Terms

Symbol	Description
t and T	The current timestep, and the whole length of an episode
p and p^*	The 2-d position and the target position of the robot
ψ and ψ^*	The heading and the target heading of the robot
$q, \dot{q}$, and $\dot{q}_{\lim}$	The joint positions, joint velocities, and the joint velocity limit of the robot
v, v^H, and ω	The linear velocity, horizontal linear velocity, and angular velocity of the robot base
z	The height of the robot base
v_f and F_f	The linear velocity and the contact force of the f-th foot
a	The action, i.e., the target joint positions
τ and $\tau_{\lim}$	The torques and the torque limit of the joints
g	The projected gravity on the robot base
$\delta_T(t_0)$	$\frac{1}{t_0} \cdot \mathbb{1}(t > T - t_0)$, the reward duration mask for the last t_0 seconds
$\delta_p(d_0)$	$\mathbb{1}(\lVert p_t - p_t{}^*\rVert < d_0)$, the robot position mask
$\delta_\psi(\theta_0)$	$\mathbb{1}(\lVert \psi_t - \psi_t^*\rVert < \theta_0)$, the robot heading mask
"S2"	The abbreviation for "Stones-2Rows"
"BB"	The abbreviation for "Balance-Beams"

Reward Term	Expression	Weight
Position Tracking	$\frac{1}{1+\lVert p_t - p_t{}^*\rVert^2}$	$10\delta_T(2)$
	for "S2" and "BB"	$25\delta_T(4)$
Heading Tracking	$\frac{1}{1+\lVert \psi_t - \psi_t{}^*\rVert^2}$	$5\delta_p(2)\delta_T(4)$
	for "S2" and "BB"	$12\delta_p(2.5)\delta_T(4)$
Termination Penalty	const for early termination	-200
Collision Penalty	const for thighs and shanks	-1
Joint Velocity Penalty	$\lVert \dot{q}_t \rVert^2$	-0.001
Joint Velocity Limit	$\sum_{i=1}^{12} \max(\lvert \dot{q}_{i,t}\rvert - 0.9\dot{q}_{\lim}, 0)$	-1
Base Accel. Penalty	$\lVert \dot{v}_t \rVert^2 + 0.02\lVert \dot{\omega}_t \rVert^2$	-0.001
Feet Accel. Penalty	$\sum_{f=1}^{4} \lVert \dot{v}_{f,t} \rVert$	-0.0005
Action Rate Penalty	$\lVert a_t - a_{t-1} \rVert^2$	-0.01
Torque Penalty	$\lVert \tau_t \rVert^2$	-0.00001
	for "BB"	-0.00002
Torque Limit	$\sum_{i=1}^{12} \max(\lvert \tau_{i,t}\rvert - \tau_{\lim}, 0)$	-0.2
	for "BB" w/ $\tau'_{\lim} = 0.8\tau_{\lim}$	-0.5
Contact Force Penalty	$\sum_{f=1}^{4} \mathrm{clip}(\lVert F_{f,t} \rVert - 700, 0, 700)^2$	-0.000025
Don't Wait	$\mathbb{1}(\lVert v_t \rVert < 0.2)$	$-(1-\delta_p(1))$
Move in Direction	$\cos\langle v_t, p_t^* - p_t\rangle$	1 for first 150 iter
Stand Still	$2.5\lVert v_t \rVert + 1\lVert \omega_t \rVert$	$-\delta_p(0.25)\delta_\psi(0.5)\delta_T(1)$
Aggressive Motion	$(\lvert v_t^H \rvert - 1)^2 \cdot \mathbb{1}(\lvert v_t^H \rvert > 1)$	0
	for "S2" and "BB"	-5
Stand Pose	$\lvert z_t - 0.6 \rvert + g_{x,t}^2 + g_{y_t}^2$	0
	for "S2" and "BB"	$-5\delta_p(0.25)\delta_\psi(0.5)\delta_T(1)$

B. Learning Settings

Some learning-related settings are presented in Table V.

参阅视频附件以观察涌现行为。值得一提的是，我们的控制器容易受到映射漂移的影响，5厘米的里程计漂移就可能导致机器人摔倒，尽管这部分归因于高度危险的地形。

六、局限性与讨论

尽管我们能够在各种地形上实现敏捷运动，但由于不同地形上的学习进度可能不同且相互干扰，我们难以训练一个适用于所有地形的统一策略。此外，我们针对不同的专家任务使用了略有不同的设置，并借助人工经验来帮助仿真到现实迁移。奖励函数的调参可能十分繁琐，且罕见的失败案例难以解释。

具体而言，任务奖励函数是多种动机的"平衡点"。首先，由于我们希望机器人能够在没有人工先验的情况下自动发现到达目标的方式，因此较短的奖励持续时间是更可取的，但这也可能导致稀疏奖励和困难的探索。其次，较长的回合长度可以使动作在仿真到现实迁移中不那么激进，但也会使任务奖励变得稀疏。因此，奖励调参十分繁琐，而我们在本文中通过预训练和微调部分规避了这些权衡。

关于失败案例，要达到100%的成功率总是困难的，罕见的失败案例确实存在。然而，与基于模型的控制器（其失败案例易于理解，例如漏步、打滑或意外碰撞）相比，我们的控制器的失败案例更难解释。在这些罕见情况下，机器人可能会在存在可行落脚点时将脚悬空放置，这可能是由于神经网络的奇异性 [35] 所致。我们相信，对比学习 [36] 等表示学习技术可能有助于解决这一问题。

七、结论与未来工作

在本文中，我们实现了在具有挑战性的危险地形上进行敏捷四足运动的端到端学习。为此，我们提出使用导航公式化以及一种在先前工作基础上进行了多项改进的探索策略。我们应用两阶段训练来克服挑战性地形上的探索难题，其中针对不同地形的专家策略是从一个通用策略微调而来的。

我们的未来工作将包括：1）基于在线表示技术（如 [23]中的方法）实现地形感知的机载化；2）训练一个适用于所有类型地形的统一策略以进行运动控制；以及3）使学习到的策略更具可解释性和鲁棒性。

附录

A. 任务规格

与任务相关的域随机化设置见表III，使用的符号和奖励项见表IV。

表III 不同任务中的域随机化设置

参数	随机化范围
Stones-Everywhere	
回合时长（秒）	$[5,7]$
目标距离（米）	$[1.5,4.9]$
Stones-2Rows	
回合时长（秒）	$[6,8]$
目标距离（米）	$[3.5,4.5]$
高度图扫描漂移（米）	$[-0.05,0.05]$
Balance-Beams	
回合时长（秒）	$[8,10]$
目标距离（米）	$[3.5,4.5]$
高度图扫描漂移（米）	$[-0.025,0.025]$
质心偏移（米）	x: $[-0.15,0.15]$ y: $[-0.05,0.05]$ z: $[-0.1,0.2]$
Stepping-Beams	
回合时长（秒）	$[5,7]$
目标距离（米）	$[2.5,4.0]$

表IV 使用的符号和奖励函数项

符号	描述
t 和T	当前时间步，以及回合总长度
p以及 p^*	机器人的二维位置和目标位置
ψ 和 ψ^*	机器人的航向和目标航向
问 ·问，以及 · 问$_{\lim}$	关节位置、关节速度以及关节速度限制
v，v^H，以及 ω	线速度、水平线速度以及角机器人基座的角速度
z	机器人基座的高度
v_f 以及F_f	第f只脚的线速度和接触力
a	动作，即目标关节位置
τ 和 $\tau_{\lim}$	关节的扭矩和扭矩限制
g	机器人基座上的投影重力
$\delta_T(t_0)$	$\frac{1}{t_0} \cdot \mathbb{1}(t > T - t_0)$，最后一段时间的奖励持续时间掩码 t_0 秒
$\delta_p(d_0)$	$1(\lVert p_t - p_t{}^*\rVert < d_0)$，机器人位置掩码
$\delta_\psi(\theta_0)$	$1(\lVert \psi_t - \psi_t^*\rVert < \theta_0)$，机器人航向掩码
"S2"	"Stones-2Rows" 的缩写
"BB"	"Balance-Beams" 的缩写

奖励项	表达式	权重
位置跟踪	$\frac{1}{1+\lVert pt - pt^*\rVert^2}$	$10\delta_T(2)$
	对于*"S2"*和*"BB"*	$25\delta_T(4)$
航向跟踪	$\frac{1}{1+\lVert \psi_t - \psi_t{}^*\rVert^2}$	$5\delta_p(2)\delta_T(4)$
	对于*"S2"*和*"BB"*	$12\delta_p(2.5)\delta_T(4)$
终止惩罚	提前终止的常数	-200
碰撞惩罚	大腿和小腿的常数	-1
关节速度惩罚	$\lVert \dot{q}_t \rVert^2$	-0.001
关节速度限制	$\sum_{i=1}^{12} \max(\lvert \dot{q}_{i,t}\rvert - 0.9\dot{q}_{\lim}, 0)$	-1
基座加速度惩罚	$\lVert \dot{v}_t \rVert^2 + 0.02\lVert \dot{\omega}_t \rVert^2$	-0.001
脚部加速度惩罚	$\sum_{f=1}^{4} \lVert \dot{v}_{f,t} \rVert$	-0.0005
动作速率惩罚	‖一个$_t$ −一个$_{t-1}$‖^2	-0.01
扭矩惩罚	$\lVert \tau_t \rVert^2$	-0.00001
	针对*"BB"*	-0.00002
扭矩限制	$\sum_{i=1}^{12} \max(\lvert \tau_{i,t}\rvert - \tau_{\lim}, 0)$	-0.2
	对于 *"BB"* w/ $\tau'_{\lim} = 0.8\tau_{\lim}$	-0.5
接触力惩罚	$\sum_{f=1}^{4} \mathrm{clip}(\lVert F_{f,t} \rVert - 700, 0, 700)^2$	-0.000025
不要等待	1（$\lVert v_t \rVert < 0.2$）	$-(1-\delta_p(1))$
按方向移动	$\cos\langle v_t, p_t^* - p_t\rangle$	前150次迭代为1
保持静止	$2.5\lVert v_t \rVert + 1\lVert \omega_t \rVert$	$-\delta_p(0.25)\delta_\psi(0.5)\delta_T(1)$
激进运动	$(\lvert v_t^H \rvert - 1)^2 \cdot \mathbb{1}(\lvert v_t^H \rvert > 1)$	0
	用于*"S2"*和*"BB"*	-5
站立姿态	$\lvert z_t - 0.6 \rvert + g_{x,t}^2 + g_{y_t}^2$	0
	对于*"S2"*和*"BB"*	$-5\delta_p(0.25)\delta_\psi(0.5)\delta_T(1)$

B. 学习设置

表V中展示了一些与学习相关的设置。

TABLE V
LEARNING SETTINGS

Configuration	Values
Actor and critic	MLP, hidden layer size (512, 256, 128)
	All observations flattened
Exteroception	25×16 heightmap, 7cm gridsize
Number of robots	4096
Policy steps per iteration	48
Clip negative rewards to 0?	No
RND M1	MLP, hidden layer size (128, 128)
RND M2	MLP, hidden layer size (256, 256)
RND reward weight	1.0
RND optimization weight	1.0
Loss for RND M1	mean L1-loss

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表V 学习设置

配置	值
演员与评论家	多层感知机，隐藏层大小（512,256,128）
	所有观测展平
外部感知	25×16高度图，7厘米网格大小
机器人数量	4096
每次迭代的策略步数	48
将负奖励裁剪为0?	No
随机网络蒸馏 M1	多层感知机，隐藏层大小（128,128）
随机网络蒸馏 M2	多层感知机，隐藏层大小（256,256）
随机网络蒸馏奖励权重	1.0
随机网络蒸馏优化权重	1.0
随机网络蒸馏 M1 的损失	平均L1损失

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